

1. DEFINITION

DEFINITION
Glycerol monostearate 40-55 (Type I) EP / Mono and diglycerides NF / Glyceryl mono- and distearate ChP

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl Stearate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	KC98R082HJ (GLYCERYL MONO AND DIPALMITOSTEARATE)
ADDITIONAL INFORMATION ON THE DENOMINATION	Glyceryl mono and dipalmitostearate C16-18 mixed mono- and diglycerides C16-18 mono- and di-glycerides Glycerides, C16-18 mono- and di- Mono- and di-glycerides [NF] Palmitostearoyl mono and diglycerides
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (Oral, topic and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Glycerol and Stearic acid 50. The esterification reaction is followed by a pelletization step. No solvent or catalyst used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Palmitostearic acids (C16 and C18), the monoester fraction being predominant (Specification: 42.3 – 51.7%).	
Full Labeling (Key letters system)	
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Glyceryl Stearate	$A_1 \geq 75.0$ %
Other constituents (i.e. incidental ingredients)	
None	



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

4. DRUG MASTER FILE

Country	Excipient master file number
US DMF (Type IV – Excipient)	/
Canada DMF (Type III – Excipient)	/
Other countries	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

CHINA BUNDLING REVIEW			
(English name) Chinese name	Registration N°	Company	CFDA/CDE Reference
Glyceryl mono- and distearate 单双硬脂酸甘油酯	F2019000559 (Route 1)	GATTEFOSSE SAS	Raw material approved for use in marketed formulations in Joint-Review with drug applications (Status A).
Glyceryl mono- and distearate 单双硬脂酸甘油酯	F20190000746 (Route 2)	GATTEFOSSE SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications. However, the excipient dossier has already been submitted (Status I).
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Glycerol monostearate 40-55 (Type I)
USP/ NF Current edition	Mono and diglycerides
Chinese Pharmacopoeia Current edition	Glyceryl mono- and distearate



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓ ⁽⁵⁾⁽⁶⁾ GRAS 184.1505		
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl palmito-stearate	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines; Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines,	/
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Glyceryl palmitostearate	<u>Non-medicinal:</u> Purposes: Diluent , Emulsifying Agent , Lubricant
Approved chemical name	Glyceryl monostearate	<u>Medicinal:</u> Rationale: Classified as an NHP under Schedule 1, item 2 (an isolate) of the Natural Health Products Regulations. Reference: NNHPD <u>Non-medicinal:</u> Purposes: Diluent , Emulsifying Agent , Flavour Enhancer , Fragrance Ingredient , Humectant , Lubricant , Solubilizing Agent , Stabilizing Agent , Surfactant - Emulsifying Agent
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	Route	Species	Test article	Results/Conclusion/Conclusions	Reference
Acute irritation	Dermal	Rabbit	GELEOL™ Pure	Cutaneous irritation index: 0.17 Non irritant	(Gattefossé data, GAT -78177, 1978)
Acute irritation	Ocular	Rabbit	GELEOL™ 30 % in Olive Oil	Ocular irritation index (acute): 12.0 Ocular irritation index (48h): 0.33 Slightly Irritant	(Gattefossé data, GAT -78177, 1978)
OECD : Organisation for Economic Co-operation and development - JORF: Journal officiel République Française (Official Journal French Republic)					
Summary of files available upon request, please contact your Gattefossé local representative. Additional literature review is available for the different endpoints, please consult the document "Tox and safety overview" for a complete review on regulatory, tox and safety aspects.					

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	Reference	Maximum "as used" concentration	Conclusions
Glyceryl stearate and Glyceryl stearate SE	JACT 1(4):169- 192, 1982	Glyceryl Stearate Up to 50%	Based on the available animal and clinical data included in the report, the Panel concludes that Glyceryl Stearate and Glyceryl Stearate/SE are safe for topical application to humans in the present practices of use and concentration.



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EC #	Chemical name
Glyceryl Stearate	85251-77-0	286-490-9	Glycerides, C16-18 mono- and di-
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACH submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Inventory names
Glyceryl Stearate	Listed	DSL listed	Listed	Listed	Glycerides, C16-18 mono- and di- (AIIC, DSL, EINECS, NZIoC, PICCS, REACH, TCSI, TSCA)/ Glycerides, C16-18 mono- and di- ester (IECSC)/ Glycerides, (C16-18) mono- and di- (AREC, ECL)
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.					
Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.					
Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					
Australia: AIIC - Canada: DLS/NDSL - China: IECSC - European Union: EINECS, REACH - Japan: ENCS - Mexico: INSQ - New Zealand: NZIoC - Philippines: PICCS - South Korea: ECL, AREC - Switzerland: SWISS - Taiwan: TCSI - Thailand: TDCA - USA: TSCA - Vietnam: VNECI					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Glyceryl Stearate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL MONO AND DIPALMITOSTEARATE (UNII : KC98R082HJ)	ORAL	CAPSULE		16 mg
	ORAL	CAPSULE, DELAYED RELEASE		25 mg
	ORAL	TABLET, DELAYED RELEASE		3 mg
	ORAL	TABLET, ORALLY DISINTEGRATING, DELAYED RELEASE		28 mg
	TOPICAL	CREAM		312 mg
	VAGINAL	CREAM		160 mg
MONO AND DIGLYCERIDE (UNII: NA)	ORAL	CAPSULE CAPSULE, DELAYED RELEASE	289.47 mg	68 mg
	TOPICAL	AEROSOL, FOAM CREAM CREAM, AUGMENTED OINTMENT	0.54 % w/w 15 % w/w 1 % w/w 7% w/w	
	VAGINAL	CREAM		160 mg
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL MONOSTEARATE (UNII: 2300U9XXE4)	AURICULAR (OTIC)	SUSPENSION	0.5%w/v	
	OPHTHALMIC	SUSPENSION	0.5%w/v	
	ORAL	CAPSULE	13.75mg	
	ORAL	CAPSULE, DELAYED RELEASE		15 mg
	ORAL	CAPSULE, EXTENDED RELEASE	32.4 mg	
	ORAL	EMULSION	NA	
	ORAL	GRANULE, FOR SUSPENSION		4 mg
	ORAL	TABLET	33.34 mg	
	ORAL	TABLET, CHEWABLE		1 mg
	ORAL	TABLET, DELAYED RELEASE		6 mg
	ORAL	TABLET, EXTENDED RELEASE		272 mg
	ORAL	TABLET, FILM COATED		5 mg
	ORAL	TABLET, FILM COATED, EXTENDED RELEASE	264.3 mg	
	ORAL	TABLET, ORALLY DISINTEGRATING, DELAYED RELEASE		23 mg
	RECTAL	SUPPOSITORY	35mg	
	SUBLINGUAL	TABLET	1.23 mg	
	TOPICAL	AEROSOL, FOAM	3.27 % w/w	
	TOPICAL	CREAM		836 mg
	TOPICAL	LIQUID	1 % w/w	
	TOPICAL	LOTION	9 % w/w	
	TOPICAL	OINTMENT	5 % w/w	
	TOPICAL	SUSPENSION	1.25 % w/w	
	VAGINAL	AEROSOL, FOAM	3 % w/w	
	VAGINAL	CREAM	17 % w/w	
GLYCERYL DISTEARATE (UNII : 73071MW2KM)	ORAL	CAPSULE, EXTENDED RELEASE	39.2 mg	
	ORAL	TABLET		100 mg
	ORAL	TABLET, EXTENDED RELEASE		13 mg
	ORAL	TABLET, ORALLY DISINTEGRATING	4 mg	
GLYCERYL PALMITOSTEARATE (UNII : GSY510183C)	INTRADERMAL	INJECTION	16 mg	
	ORAL	SUSPENSION	150 mg/ 5 ml	
	ORAL	TABLET	18mg	
	TOPICAL	AEROSOL	2.55 %w/w	
	TOPICAL	CREAM	2.78 %w/w	



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
	TOPICAL	EMULSION	2.55 % w/w	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.3. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfarb)			
DENOMINATION	FUNCTIONAL CATEGORY	REGULATORY STATUS	SAFETY
Glyceryl Monostearate	Emollient; emulsifying agent; solubilizing agent; stabilizing agent; sustained-release agent; tablet and capsule lubricant.	GRAS listed. Included in the FDA Inactive Ingredients Database (oral capsules and tablets; ophthalmic, otic, rectal, topical, transdermal, and vaginal preparations). Included in nonparenteral medicines licensed in the UK. Included in the Canadian List of Acceptable Non-medicinal Ingredients.	Glyceryl monostearate is widely used in cosmetics, foods, and oral and topical pharmaceutical formulations, and is generally regarded as a nontoxic and nonirritant material. LD50(mouse, IP): 0.2 g/kg

COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS (Garg, S. et al., 2001)		
DENOMINATION	CATEGORY (CONCENTRATION IN %)	REGULATORY STATUS
Glyceryl monostearate	emulsifier; stabilizer; emollient; plasticizer; tableting aid	G, 1, 3, 8–10, 12–14, 23, 25, 26
Glyceryl stearate	emulsifying agent; stabilizer; emollient; plasticizer; tablet lubricant; sustained-release tablets	G, 12, 25
Glycerides of fatty acids	Suppository bases	1, 3–14, 16–26
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopelias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portugese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian		

10.2. Food/Nutraceutical/Dietary supplement applications

Mono- and Diglycerides have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

COUNTRY	REFERENCE	REGISTRATION NUMBER
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS471 Mono- and Diglycerides
E.U.	EFA (European Food additives)	E471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides



Regulatory data sheet (RDS) – Pharmaceutical market

GELEOL MONO AND DIGLYCERIDES NF (Code: 5749)

Last update: 06 November 2023

COUNTRY	REFERENCE	REGISTRATION NUMBER
U.S.A.	GRAS Generally recognized as safe according to 21 CFR	21CFR §184.1505 Mono- and Diglycerides
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

HANDBOOK OF FOOD ADDITIVES		
ASH Michael et al. (2008), <i>Handbook of Food Additives CD-ROM</i> , 3rd ed., Synapse Information Resources Inc.		
DENOMINATION	USE	SAFETY
GELEOL Glyceryl stearate	Emulsifier, consistency agent, opacifier for foods	EP, USP/NF compliance; E471, FCC, PSFA compliance; DMF no. 9052

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.

Liability clause

The information contained in this document is based on our knowledge of the related product(s) at the date indicated herein, and is provided in good faith. Notwithstanding the foregoing, no guarantee is made in respect of the product's(s') properties or efficacy, or the continued exactitude and completeness of the information herein. No contractual tie is formed by the provision of this information. This document does not relieve the user from conducting all tests and verifications to ensure the product's(s') suitability for each intended use. Nothing herein shall be construed as a recommendation or licence to use any information that does or may conflict with any intellectual property. We make no representation or warranty, express or implied, that use of this information will not infringe any intellectual property of Gattefossé or any third party. The user assumes all risks and liability incurred if a product is used other than for the purpose designated by Gattefossé. This statement does not relieve the user from verifying its conformity with all current applicable laws and regulations, and the user shall remain solely liable for ensuring its continued conformity therewith.

